Fiscal Year:	FY 2019	**Task Last Updated:**	FY 03/04/2019
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Simulation of GCR-Induced Harderian Gland and Lung Tumorigenesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	eablakely@lbl.gov	**Fax:**	FY 510-486-4475
PI Organization Type:	UNIVERSITY	**Phone:**	510-486-6595
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	One Cyclotron Road		
PI Address 2:	MS977		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	02/04/2016	**End Date:**	02/04/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is 2/4/2021 per PI (Ed., 11/7/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mao, Jian-Hua Ph.D. (Lawrence Berkeley National Laboratory) Chang, Polly Ph.D. (SRI International)		
Grant/Contract No.:	NNJ16HP22I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Increased cancer risk remains one of the primary concerns for travel into deep space and may preclude manned missions to Mars due to the large uncertainties that currently exist in estimating cancer risk from the spectrum of radiation types found in space with available human epidemiological data on radiation-induced cancer. Existing data on human risk of cancer from exposures to X- and gamma-rays must be scaled to the many types and fluxes of radiations found in space using radiation quality factors and dose-rate modification factors. Linearity of dose-response is often assumed since the shape of the dose-response at doses below 100 mSv is unknown. However, preliminary analysis of our recently completed experiments with silicon and titanium ion suggest non-linear structure in the low-dose region for both Harderian and lung tumorigenesis. The goal of this proposal is to reduce uncertainties in the estimation of particle radiation carcinogenesis from galactic cosmic radiations (GCR). Our approach is to use the existing data on Harderian gland tumorigenesis from investigations of individual high-atomic number, and high-energy (HZE) ion beams to predict outcome of mixed ion chronic GCR exposures, and then measure Harderian gland and lung dose-dependent tumor data for such exposures. We will compare some of the measured cancer prevalence outcomes of a GCR-simulated exposure with the theoretical predictions based on experiments with individual ion beams. Our main hypothesis is that HZE carcinogenic effects are LET- (linear energy transfer) and dose-dependent and not additive at low particle doses due to the contribution of non-targeted effects. Completion of the proposed GCR studies using a mouse strain that has already been well characterized with a low spontaneous tumor background will allow a significant test of modeling approaches used to estimate carcinogenesis risk from a simulated GCR beamline.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Completion of our research on predicting and measuring HZE tumorigenic effects to test the hypothesis that HZE carcinogenic effects and LET- and dose-dependent are not additive at low particle doses due to the contribution of non-targeted effects will allow a significant test of modeling approaches used to estimate carcinogenic risk from single-ion and mixed radiation fields. This work will also benefit estimation of risk of tumorigenesis from mixed radiation fields on Earth.
Task Progress:	Technical Progress: Recent experimental milestones 1--Participated in NASA Space Radiation Laboratory (NSRL)18A (2-3 May 2018)—Exposed a total (with unirradiated controls) of 180 female CB6F1 mice to a 6-ion mixed beam delivered in rapid sequence to a total dose of 100 cGy within 24 minutes. The ions, beam energies, LET values, and dose fractions were: 250 MeV/u protons (0.4 keV/µm-60 cGy), 228 MeV/u helium (1.6 keV/µm-20 cGy), 350 MeV/u oxygen (17 keV/µm-10 cGy), 260 MeV/u silicon (70 keV/µm-2.5 cGy), 1 GeV/u titanium (100 keV/µm-2.5 cGy), and 600 MeV/u iron (190 keV/µm-5 cGy). This was an historic first test of the “automatic” computer-driven control of the rapid-sequence, multi-ion beam delivery system for mixed-beam and GCR-simulation exposures recently developed by Dr. Adam Rusek and his team. The total dose of 100 cGy was delivered at an average dose rate of ~10 cGy/min). Dr. Megumi Hada and her team (Andrew Beitman and Jordan Rhone) completed 9 cave entries exposing human blood, and human epithelial cells and fibroblasts. Dr. Hada exposed sets of her samples to each of the 6-ion beams alone, followed by the 6-ion sequence at the dose fractions to be used for the mice totaling 100 cGy, as well as half and quarter proportional total doses of 50 cGy and 25 cGy. The Blakely/Chang group followed with the irradiation of 136 female, CB6F1 mice of 100-120 days of age in 7 hours on May 2nd, 2018 from 1:06pm to 8:00pm. Each animal received a sequence of 60 cGy 250 MeV protons, 20 cGy 228 MeV/u He, 10 cGy 350 MeV/u Oxygen, 2.5 cGy 260 MeV/u Si, 2.5 cGy 1,000 MeV/u Ti, and 5 cGy 600 MeV/u Fe within an approximately 24 min period. Eight mice were whole-body irradiated in each of 17 cave entries. The automated delivery of mixed ion beams by Adam Rusek and his team at Brookhaven National Laboratory (BNL) represents a major leap forward in NSRL capabilities. 2—We harvested tumors and tissues from two large animal cohorts totaling 400 mice in 2018. In February, we necropsied animals from the single or dual-ion beam studies from NSRL16C, and in August, from the 350 MeV/u Oxygen ion dose response of NSRL17C. Both cohorts included harvesting of murine bone marrow cells for later chromosome analysis. This was an enormous effort, and was successfully completed on time and on budget. The results were quite striking. CHROMOSOME ABERRATIONS (DUAL BEAMS). Murine chromosome aberrations in the bone marrow of mice harvested at 16 months post irradiation were analyzed. Animals were either sham-treated (controls) or irradiated with 60 cGy of protons (H) alone, 40 cGy of Silicon ions alone, or 60 cGy of protons followed by 40 Gy of silicon ions, 40 cGy of protons followed by 30 cGy of iron ions, or 20 cGy of silicon ions followed by 20 cGy of iron ions. The number of simple and complex exchanges were measured in each of 4 - 5 mice in group. One of the key observations was the significant increased heterogeneity in chromosome exchanges/100 cells in the mice exposed to dual ion beams. Both simple and complex exchanges increase after exposure to HZE ions, as well as the variability between individual mice. These observations will require follow up in order to understand any potential correlations with tumor outcome. OXYGEN ION HG TUMORIGENESIS DOSE RESPONSE. The 350 MeV/u Oxygen ion Harderian Gland tumorigenesis prevalence dose response data were shown to nest between the 670 MeV/u Neon ion data (LET of ~25 keV/µm and the 228 MeV/u Helium ion data (LET of ~1.6 keV/µm). These preliminary data are consistent with our expectations. A manuscript is in preparation. 3--In Silico Modeling of murine Harderian Gland tumorigenesis. It is currently unknown whether the GCR mixed field could demonstrate statistically significant synergy among its components. Synergy would increase risks for prolonged astronaut voyages in inter-planetary space. Prof. Ray Sachs and his team of 7 undergraduate Mathematics students at the University of California, Berkeley recently modeled the NSRL dual-beam HG tumor prevalence data. The methods were those introduced or summarized in the team's Radiat.Environ Biophys paper by Huang et al., 2019. These methods included the following: • Using their incremental effect additivity (IEA) synergy theory based on computer numerical integration of non-linear ordinary differential equations. • Analyzing IEA no-synergy/no-antagonism baseline dose-effect relation uncertainties by Monte Carlo sampling from variance-covariance matrices, thereby taking adjustable parameter correlations appropriately into account and getting appropriately narrow 95% CI. • Taking advantage of the IEA mix-mix principle to transfer uncertainties from matched one-ion analyses to mixture analyses, even when nominally one-ion NSRL beams are actually mixtures at the target due to intervening matter. The single total dose dual beam Harderian Gland tumorigenesis data points we observed in our three separate dual-beam combinations have been compared to calculated incremental effect additivity synergy theory results with 95% CI. With

	the 60 cGy protons + 40 cGy silicon, and also the 20 cGy Si + 20 cGy Fe study, the observed mixture data points were slightly above the red IEA baseline representing no synergy or antagonism and extended beyond the CI, therefore the test of our hypothesis indicates there is no statistically significant synergy in these two cases. However, in the case of 40 cGy protons + 30 cGy Fe, unexpected significant synergy between proton and iron ion action is observed. Additional theoretical modeling of the yield of human chromosome aberrations from single and mixed ion beams at NSRL is ongoing in collaboration with Dr. Megumi Hada. 4—Ongoing and upcoming plans: We plan to irradiate 2 total dose points (30 cGy, 60cGy) of a rapid sequence 3 HZE (260 Mev/u Si, 1 Gev/u Ti and 600 MeV/u Fe) beam mixture each contributing equal 10 cGy or 20 cGy dose fractions during NSRL19A in April 2019. This animal study of 420 mice is an important test of our hypotheses regarding the role of HZE non-targeted effects on tumorigenesis, and will bring our animal inventory up to 600. Mice irradiated in May 2018 during NSRL18A with a 6-ion mixed beam rapid sequence will be necropsied in November 2020 to harvest tumors, and to harvest bone marrow for chromosome aberration analysis. Reference Huang, EG, Lin, Y, Ebert M, Ham DW, Yunzhi C, and Sachs RK, Synergy theory for murine Harderian gland tumours after irradiation by mixtures of high-energy ionized atomic nuclei, Radiation and Environmental Biophysics. https:// . 2019 Feb 2.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Abstracts for Journals and Proceedings	Chang PY, Bakke J, Grover A, Rusek A, Rosselot R, Synder D, Rosen C, Bjornstad KA, Beitman AJ, Rhone JR, Mao JH, Hada M, Sachs RK, Blakely EA. "Harderian gland tumorigenesis: Dual particle beam interactions." 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. Radiation Research 2018 Abstract Book. , Sep-2018
Abstracts for Journals and Proceedings	Beitman AJ, Rhone JR, Chang PY, Blakely EA, Sachs RK, Saganti PB, Hada M. "Chromosome aberrations induced by GCR stimulated mixed beam exposure in human lymphocytes, fibroblasts and epithelial cells." 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. Abstract book. , Jan-2019
Abstracts for Journals and Proceedings	Goukassian D, Cimini M, Garikipati S, Addya S, Blakely E, Chang P, Kishore R. "Cardiovascular disease risks due to single full body gamma and HZE radiation--Long term dose response studies." 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. Abstract Book. , Jan-2019
Abstracts for Journals and Proceedings	Blakely EA. "Current approaches to evaluate space radiation health risks and mitigation." Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. Abstract Book. , Jul-2018
Abstracts for Journals and Proceedings	Blakely E, Chang P, Bakke J, Grover A, Rusek A, Snyder D, Rosen C, Bjornstad K, Beitman A, Rhone J, Mao J, Hada M, Sachs R. "Harderian gland tumorigenesis: Dual particle beam exposures." 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. Abstract Book. , Jan-2019
Abstracts for Journals and Proceedings	Huang E, Ham DW, Lin Y, Wang S, Zhao A, Ebert M, Blakely E, Chang P, Sachs R. "Synergy theory: murine Harderian gland tumors and in vitro chromosome aberrations induced by exposure to mixed beams with some high-LET components." 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. Abstract Book. , Jan-2019
Articles in Peer-reviewed Journals	Ham DW, Song B, Gao J, Yu J, Sachs RK. "Synergy theory in radiobiology." Radiation Research. 2018 Mar;189(3):225-37. Epub 2017 Dec 29. https://doi.org/10.1667/RR14948.1 ; PMID: 29286257 , Mar-2018
Articles in Peer-reviewed Journals	Huang EG, Lin Y, Ebert M, Ham DW, Yunzhi C, Sachs RK. "Synergy theory for murine Harderian gland tumours after irradiation by mixtures of high-energy ionized atomic nuclei." Radiat Environ Biophys. 2019 May;58(2):151-66. https://doi.org/10.1007/s00411-018-00774-x ; PMID: 30712093 , May-2019
Awards	Blakely EA. "Named a Distinguished Emeritus Member of the NCRP (National Council on Radiation Protection and Measurements) at 2018 Annual Meeting of the NCRP, Bethesda, MD, 5-6 March 2018." Mar-2018